MERKNET

Blockchain-Based Cross-Border Trade Documentation
and Execution Layer for Indian Manufacturing Exports

Submission to the Blockchain India Challenge

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Problem Statement

> *Blockchain technology can fundamentally transform cross-border digital trade by creating a decentralized, trusted and globally interoperable layer for the issuance, exchange and verification of critical documents such as e-Bills of Lading, Certificates of Origin, Inspection Certificates etc. — C-DAC, Blockchain India Challenge*

Cross-border manufacturing sourcing from India fails not because of bad suppliers, but because no intermediary owns the outcome. The documentation infrastructure is fragmented, manual, and trust-deficient. From Merkantis's direct experience executing 200+ cross-border component orders across India, Europe, and the Middle East:

- **6–8 stakeholders per order, zero shared records.** Manufacturer, inspector, freight forwarder, customs broker, bank, insurance, and buyer each maintain separate documentation. No shared ledger exists.
- **Weeks of document processing.** Manual verification of Certificates of Origin, material test reports, and Bills of Lading delays clearance significantly. Much of this delay stems from signature verification — government agencies and banks must independently confirm that each document was signed by an authorized party, a process that today involves phone calls, emails, and physical stamp checks.
- **Frequent document discrepancies.** Indian MSME export shipments regularly face discrepancies at destination ports, leading to demurrage charges and disputes. Discrepancies often arise because there is no way to verify, at the point of origin, that a Certificate of Origin or inspection report carries a legitimate signature.
- **No verifiable chain of custody.** Quality certificates and inspection reports are PDF files passed via email. Forgery is trivial — a digital signature on a PDF can be visually spoofed, and verification requires manually contacting the issuing authority. There is no cryptographic binding between the signer's identity and the document.
- **Duplicate financing fraud.** The same Bill of Lading can be presented to multiple banks for trade finance because no shared, tamper-proof record of document ownership exists.
- **Communication asymmetry.** Suppliers quote optimistic timelines with no shared, immutable record of commitments vs. actuals.

India already has the cryptographic infrastructure to solve the signature problem: Digital Signature Certificates (DSCs) issued by CCA-accredited authorities under the IT Act 2000 are X.509 certificates — the same standard used by Hyperledger Fabric for identity. The missing

piece is a shared ledger where these signatures can be verified by any stakeholder, instantly, without contacting the issuing authority. That is what MerkNet provides.

Innovative Solution

MerkNet is a permissioned blockchain platform that digitizes the entire lifecycle of cross-border manufacturing trade — from RFQ issuance through final delivery and payment — as an immutable, shared record across all stakeholders. Built on Hyperledger Fabric 3.0+ with SmartBFT consensus, FireFly middleware, and IPFS document storage.

Core Innovation Layers

1. Progressive Decentralization. Merkantis starts as a trusted principal; new participants get a managed identity with zero infrastructure. Merkantis holds their Fabric CA credentials and signs on their behalf — but every transaction is still attributed to them on-chain. After the first successful deal, participants may transition to self-sovereign nodes (own peer, own keys). Transition criteria: at least one completed deal, KYC verified, and organizational readiness assessment. Key migration is handled by Fabric CA re-enrollment; transaction history attributed to the managed identity remains on the ledger and is linked to the new self-sovereign identity via the MSP. This eliminates the cold-start problem that killed TradeLens, while creating a clear path to full decentralization.

2. Full Document Lifecycle On-Chain. Every state change from RFQ to delivery is a Fabric transaction. Documents are hashed on-chain and stored encrypted on IPFS. Critically, when a DGFT officer issues a Certificate of Origin, they sign the transaction with their DSC — that signature is cryptographically verifiable by any peer on the network, instantly, without contacting DGFT. The same applies to inspection certificates, customs declarations, and Bills of Lading. This replaces today's manual signature verification with cryptographic proof.

3. On-Chain Reputation. Replaces escrow with permanent accountability. Deal milestones are hashed to the network channel; aggregated scores (payment timeliness, quality acceptance, delivery reliability) determine access tier.

4. Cryptographic Privacy. Per-deal AES-256 envelope encryption via ECIES ensures channel peers cannot read deals they're not party to. UCANs handle document-level authorization with delegation chains.

5. Compliance Engine. Automated validation against DGFT policies, destination regulations (CE/REACH/RoHS), and bilateral agreements. Runs as chaincode; rules updatable by DGFT/Customs without chaincode upgrades.

6. Payment Tracking. Chaincode records milestones, emits events on completion via FireFly. Late payments hit the public ledger. No funds held on-chain.

7. Supply Chain Finance. OCEN-aligned financing channel lets suppliers present verified on-chain POs to banks. e-BLs minted as non-fungible tokens eliminate duplicate financing fraud.[1]

8. QR-Code Verification. Sub-second hash lookups against the local peer. Works in low-connectivity factory and port environments.

9. Modular Integration. FireFly REST APIs integrate with existing ERP, TMS, banking, and customs systems. MerkNet is an interoperability layer, not platform lock-in.

[1]See Appendix D (Reputation and SupplyChainFinance chaincodes) for full state machines and verification mechanisms.

Blockchain Applicability

Why Not a Centralized Database?

Merkantis itself is a stakeholder in every trade it facilitates. Buyers and suppliers should not have to trust Merkantis's database for the integrity of their documents — just as they should not have to trust any single bank, customs authority, or logistics provider. A permissioned blockchain eliminates this requirement: each party runs or accesses a peer that independently validates every transaction. Merkantis operates the network but cannot unilaterally alter records.

A centralized database with multi-party write access (like ICEGATE for customs) solves part of the problem, but only within a single jurisdiction and a single workflow. Cross-border trade involves Indian suppliers, European buyers, banks in both jurisdictions, customs in multiple countries, and independent inspectors. No single database operator commands trust across all of these.

The Signature Verification Argument

Today, government agencies verify Certificates of Origin, inspection certificates, and Bills of Lading by checking signatures — often manually. A DGFT officer signs a CoO; a customs agent at the destination port must verify that signature by contacting DGFT or relying on intermediary attestations. This is slow, error-prone, and the primary source of document processing delays. India's existing DSC infrastructure is X.509-based and cryptographic. On a Fabric network, a DGFT officer's DSC becomes their transaction-signing identity. When they issue a CoO on-chain, the signature is bound to the document hash and verifiable by any peer — a bank in Mumbai, a customs agent in Hamburg, or a buyer in Zurich — in sub-second time, with no phone calls. The blockchain is not adding a new identity layer; it is making India's existing digital signature infrastructure interoperable across borders.[2]

Blockchain Property	Pain Point	Resolution
Cryptographic Signing	Manual signature verification for CoO, BL, inspection certs	DSC-signed on-chain transactions verifiable by any peer instantly
Decentralization	No single party trusted by all 6–8 stakeholders	Shared ledger; no reliance on any intermediary

[2]India's DSC infrastructure is X.509-based and issued by CCA-accredited authorities (eMudhra, Sify, NCode) under the IT Act 2000. These certificates are directly compatible with Fabric CA's external CA support, requiring no additional identity layer.

Immutability	Documents altered post-issuance; certificates backdated	Permanent, timestamped Fabric blocks
Smart Contracts	Manual compliance checks; payment disputes	Automated milestone tracking and validation
Non-repudiation	Commitments denied; timelines revised retroactively	Cryptographically signed, undeniable transactions
Auditability	Weeks to reconstruct an order's history	Regulators query their own peer directly
Privacy	Commercial terms must stay confidential	Envelope encryption; hashes visible, data hidden

Technical Approach and Architecture

MerkNet's architecture is summarized below. Full network topology, channel details, privacy layer, identity model, and end-to-end data flows are in the appendix.[3]

Layer	Technology	Role
Consensus	SmartBFT (Fabric 3.0+)	BFT ordering across independent orgs
Ledger	Hyperledger Fabric	Transactions, chaincode, world state
Privacy	PDCs + Envelope Encryption + UCANs	Per-deal cryptographic data isolation
Identity	Fabric CA with ABAC + DSC support	X.509 certificates, role attributes, govt DSCs
Middleware	Hyperledger FireFly	REST API, event bus, token management
Documents	IPFS (encrypted, pinned)	Content-addressed, peer-distributed
Application	Merkantis Platform	Web/app interface, QR verification

Three-Channel Design

Network Channel — identity registry, reputation scores, compliance rules, and milestone history (hashes only, no commercial data). All participants including banks.

Trade Channel — all deal lifecycle transactions. The ledger stores only hashes and encrypted blobs; plaintext is never on-chain.

Financing Channel (Phase 2+) — banks and financial institutions. Loan terms separated from trade data. Banks verify documents by hashing against the network channel.

Phased Deployment

Phase 1 (MVP): Merkantis + DGFT + Customs = 4 orderers (F=1). All suppliers/buyers via managed identities.

Phase 2: Large participants get own peers. 7 orderers (F=2). Merkantis HA.

[3]See Appendix A for full network topology, channel architecture, privacy layer, identity model, and end-to-end data flows.

Phase 3: Major orgs run own infrastructure. Merkantis transitions from principal to platform operator.

Benchmarking and Differentiation

Every previous attempt at blockchain-based trade platforms has shut down or remained narrowly scoped. MerkNet learns from each failure.[4]

Dimension	Previous Attempts	MerkNet
Scope	Shipping only (TradeLens) or finance only (Marco Polo)	Full lifecycle: RFQ → delivery → payment
Status	TradeLens defunct 2022; Marco Polo defunct 2023	Proposed, grounded in 200+ live engagements
Manufacturing	None covered quality/inspection	QC workflow, inspection anchoring, factory data
Onboarding	Required own nodes or bank mediation	Managed identity; zero infrastructure day one
Signatures	No integration with national PKI	DSC-native; govt officers sign with existing credentials
Privacy	Channel isolation (all members see all)	Envelope encryption: per-deal cryptographic isolation
India	No solutions beyond narrow govt pilots	200+ India engagements; DGFT/Customs integration
Finance	Bank-to-bank or none	OCEN-aligned PO/invoice verification for MSME lending

Why Merkantis

1. Live operational data from 200+ cross-border engagements — 200+ completed or in-progress procurement cycles involving supplier qualification, order management, quality inspection, and delivery across Indian manufacturing clusters. The platform digitizes these existing workflows, not theoretical ones.
2. Existing supplier network in Coimbatore, Bangalore, Chennai — pilot-ready on day one.
3. Active buyer relationships in Europe (Switzerland, Germany) — demand-side validation.
4. Domain expertise in precision manufacturing documentation, quality, and compliance.

[4]See Appendix B for detailed analysis of TradeLens, Marco Polo, CargoX, Indian government pilots, and Raymond x Spydra.

5. Cross-border entity (India + Switzerland) — firsthand regulatory understanding on both sides.

Non-Crypto Blockchain Platform

MerkNet is built entirely on non-crypto, permissioned infrastructure. No cryptocurrency, no tokens for value transfer, no mining.[5]

Hyperledger Fabric 3.0+

Permissioned enterprise ledger. SmartBFT consensus provides Byzantine fault tolerance for multi-org, government-involved deployments. Execute-order-validate model ensures orderers never see plaintext data. Chaincode in Go. De facto standard for Indian government blockchain (NIC pilots, NITI Aayog). Critically, Fabric CA's external CA support means government participants can authenticate using their existing DSCs — no new identity infrastructure required.

Hyperledger FireFly

Self-hosted middleware: REST API gateway, real-time event bus, token connector for e-BLs as non-fungible tokens without cryptocurrency, and off-chain data exchange with on-chain proof anchoring. DSC-based signing integrates via FireFly's pluggable identity layer.

IPFS

Content-addressed distributed storage. No cryptocurrency (Filecoin is separate and not used). Envelope encryption ensures only deal parties can decrypt. UCANs provide capability-based authorization with delegation chains.

[5]See Appendix C for full technology selection rationale and alternatives considered (Corda, Besu, Sawtooth, Cosmos, Indy).

Permissioned Blockchain and Smart Contracts

Every participant is an authenticated organization with an MSP identity issued by Fabric CA. No anonymous participation.[6]

Network Participants

Organization	Role	Infrastructure
Merkantis	Network operator, trade execution	Orderer(s) + peer(s) + FireFly + IPFS
DGFT	Export licensing, compliance	Orderer + peer + FireFly
Customs (CBIC)	Clearance, filing	Orderer + peer + FireFly
Suppliers	Component manufacturers	Managed or own peer
Buyers	Procurement entities	Managed or own peer
Quality Inspectors	Third-party verification	Managed identity
Logistics (Phase 2+)	Freight, shipping	Own peer
Banks (Phase 2+)	Finance, PO/invoice verification	Orderer + peer + FireFly

Progressive Identity Model

New participants start with a managed identity. After the first successful deal, they may transition to self-sovereign identity. Government participants with existing DSCs use them directly via Fabric's external CA support — their DSC becomes their signing identity on the network. When a DGFT officer issues a Certificate of Origin, the transaction is signed with their DSC and cryptographically verifiable by any peer.

Seven Chaincode Packages

DealLifecycle (RFQ to completion), QualityVerification (inspection workflow), DocumentManagement (documents as tokens, including e-BLs), PaymentTracking (milestone recording), Compliance (rule-based regulatory validation), Reputation (aggregated scores), and SupplyChainFinance (OCEN-aligned loan lifecycle).

[6]See Appendix D for the complete smart contract suite: all seven chaincode packages, state machines, endorsement policies, and multi-party signature thresholds.

Endorsement policies are multi-org: PO confirmation requires buyer + supplier + Merkantis; inspection requires inspector + buyer; customs filing requires Merkantis + customs. High-value transactions trigger internal approval thresholds via a ThresholdPolicy chaincode private to each org.

Regulatory Framework

MerkNet is designed for regulatory compliance from day one. Key frameworks summarized below.[7]

Indian Compliance

- **IT Act, 2000:** Electronic records (S.4), digital signatures (S.5), evidence admissibility (S.65B). DSCs issued under this Act are the identity foundation for MerkNet.
- **Bills of Lading Act, 2025:** Full legal recognition of blockchain-based e-BLs. MerkNet's implementation aligns directly.
- **Foreign Trade Policy 2025:** Explicitly includes blockchain security for trade records.
- **DPDP Act, 2023:** Data minimization (only hashes on ledger); key destruction as erasure; cross-border via blacklist approach.
- **RBI Digital Lending Directions, 2025:** Financing channel designed to comply. Merkantis intends to position as a Loan Service Provider under OCEN 4.0; this requires regulatory alignment and bank partnerships that are part of the Phase 2 roadmap, not current status.
- **Customs Act / ICEGATE:** FireFly REST API integration; compliance chaincode validates before submission.
- **TReDS:** Complementary — TReDS for receivables, MerkNet for PO-based pre-shipment finance.

International Compliance

- **GDPR:** On-chain data limited to salted cryptographic hashes. All PII stored encrypted off-chain. Key destruction renders ciphertext permanently unreadable — the EDPB Guidelines 02/2025 acknowledge this approach, though the regulatory position continues to evolve. Clear controller/processor agreements between Indian and European entities.
- **UNCITRAL MLETR:** e-BL follows MLETR principles for interoperability with adopting jurisdictions.
- **ICC / WCO / WTO TFA:** Document schemas aligned; digital single window implementation per Article 10.

[7]See Appendix E for the full regulatory framework covering IT Act, Bills of Lading Act 2025, DPDP Act, FEMA, RBI Directions, GDPR, MLETR, ICC/WCO standards.

Security, Privacy, Scalability and Performance

Layered, defense-in-depth security model. Full resource requirements and infrastructure costs in the appendix.[8]

Security

- **SmartBFT consensus:** Tolerates F Byzantine nodes (3F+1 orderers). Phase 1: F=1. Phase 2: F=2. No single org controls ordering.
- **DSC-native identity:** Government participants authenticate with existing Digital Signature Certificates. All transaction signatures are cryptographically verifiable by any peer.
- **Mutual TLS:** All communication mutually authenticated via X.509/DSC certificates.
- **Execute-order-validate:** Orderers never see plaintext transaction data.
- **HSM-backed keys:** Orderer and CA root keys in hardware security modules.
- **Immutable audit:** Every state change committed to all channel peers. Regulators audit from their own node.

Privacy

Two-part model: envelope encryption (AES-256 per deal, distributed via ECIES) for confidentiality; UCANs for authorization (capability-based IPFS access with delegation chains). Peers on the same channel see hashes but cannot read deals they're not party to. Privacy enforced by cryptography, not network topology.

Performance

Metric	Value
Fabric SmartBFT capacity	~3,000 TPS
Peak demand at 1,000 concurrent orders	~5–10 TPS
Headroom	~300–600x

[8]See Appendix F for detailed security architecture, resource requirements per participant type, and infrastructure cost estimates.

Document / QR verification	Sub-second (hash lookup against local peer)
IPFS retrieval (pinned content)	< 5 seconds

Execution Plan

The execution plan maps directly to the Blockchain India Challenge stages: Ideation, Prototype, MVP, and Deployment.

Stage 1: Ideation (Current)

This proposal. Problem definition grounded in 200+ operational engagements. Architecture designed, technology stack selected, regulatory mapping complete. Key output: this white paper and CDAC submission.

Stage 2: Prototype (Months 1–4)

- Deploy minimal Fabric 3.0+ network: Merkantis orderer + peer + FireFly + IPFS.
- Implement DealLifecycle and DocumentManagement chaincodes with DSC-based signing.
- Demonstrate end-to-end flow: RFQ → PO → inspection → e-BL issuance on a test network.
- Validate DSC integration: government officer signs a CoO on-chain, verified by a buyer's peer.
- QR-code verification prototype for factory and port environments.

Key output: working prototype demonstrating cryptographic document signing and verification across 2–3 stakeholder roles.

Stage 3: MVP (Months 5–9)

- Add DGFT and Customs nodes to the network (if government commitment secured; see Risks below). If government nodes are not yet available, MVP runs with Merkantis simulating government roles on separate peers with distinct MSP identities, preserving the multi-org endorsement model.
- Implement QualityVerification, PaymentTracking, and Compliance chaincodes.
- Build Merkantis web/app interface with managed identity onboarding.
- Pilot with 5–10 live cross-border orders from Merkantis's existing pipeline.
- ICEGATE integration feasibility study and initial API mapping.

Key output: MVP processing real orders with measurable reduction in document processing time.

Stage 4: Deployment (Months 10–18)

- Onboard DGFT and Customs as orderer/peer nodes (formal MoU or pilot agreement).
- Launch Financing Channel with 1–2 bank partners; implement SupplyChainFinance chaincode.
- Self-sovereign identity transitions for established participants.
- Scale to 50–100 concurrent orders; expand orderer set to 7 (F=2).
- State-level rollout planning: Tamil Nadu, Karnataka, Maharashtra.

Key output: production network processing live cross-border orders with government participation.

Expected Outcomes and Pilot Deployment

Target Metrics

Metric	Current (Manual)	Target
Document verification	3–5 days	< 5 seconds
Export clearance (platform-side)	30–45 days	< 24 hours
Payment release post-milestone	15–30 days	< 24 hours
Document discrepancy rate	15–20%	< 2%
Supplier onboarding	2–4 weeks	Hours
Dispute resolution	Weeks of email	Days
First-submission acceptance	~80–85%	98%+
Audit compliance per order	Days	Minutes

Pilot Design

The MVP pilot runs on 5–10 live cross-border orders from Merkantis's existing India-Europe pipeline: precision manufacturing components sourced from Coimbatore and Bangalore for European buyers. Pilot participants: 2–3 Indian suppliers, 1–2 European buyers, 1 third-party inspector, and — subject to government engagement — DGFT and Customs.

If government nodes are not available for the pilot, the system operates with Merkantis running separate peers with distinct MSP identities for the government roles. The endorsement policies and ABAC model remain identical; what changes is who operates the infrastructure, not what the chain validates. This allows the pilot to demonstrate the full document lifecycle, including compliance chaincode execution, without waiting for government onboarding.

Success criteria: measurable reduction in document processing time vs. Merkantis's current manual baseline, zero document discrepancies on pilot orders, and positive participant feedback on usability and trust.

Government Department Collaboration

MerkNet is designed as public-private infrastructure. Government departments participate as network nodes, strengthening consensus and gaining direct, real-time audit access.

Department	Role in MerkNet
DGFT	Primary partner. Orderer + peer. Officers sign CoOs with DSCs on-chain. Compliance chaincode validates DGFT policies; rules updatable without chaincode upgrades.
CBIC (Customs)	ICEGATE integration. QR-based form pre-population for CHAs. Orderer + peer for direct audit. Customs officers sign clearance transactions with DSCs.
Ministry of MSME	Serves 63M+ MSMEs. Zero-infrastructure onboarding. Financing channel for MSME working capital.
MeitY	Challenge organizer. MerkNet contributes to the National Blockchain Technology Stack.
RBI	Financing channel designed within Digital Lending Directions 2025. Complements TReDS.
State Industries	Pilot in Coimbatore, Bangalore, Pune. Alignment with District Export Action Plans.

Relevant Schemes

- **TIES:** Grant-in-aid up to ₹20 crore for digital trade facilitation infrastructure.
- **Districts as Export Hubs:** MerkNet as digital backbone for 734 identified districts.
- **Foreign Trade Policy 2025:** Direct implementation of its blockchain-for-trade vision.

Wider Proliferation Strategy

1. State-by-state rollout starting with manufacturing-heavy states (TN, KA, MH, GJ) through state industry partnerships.
2. Integration with existing government platforms — ICEGATE, GeM, Udyam — via FireFly APIs.
3. Industry association partnerships (CII, FICCI, CODISSIA) for supplier onboarding at scale.
4. Multi-corridor expansion: India-EU → India-UAE → India-US → India-Africa.

5. Open API ecosystem: third-party developers build inspection apps, logistics integrations, finance products.

Additional Details

Business Model

- **Transaction fee:** Per-order for blockchain anchoring and smart contract execution. Target: ₹2,000–10,000 per order depending on complexity and document count.
- **Subscription:** Monthly access by volume tier for suppliers/buyers with recurring orders.
- **Managed infrastructure:** Cloud instances for self-sovereign participants (₹5,000–15,000/month).
- **Analytics:** Aggregated supplier performance data and predictive benchmarking (subscription).
- **Finance facilitation:** Banks pay per-verification or subscription for on-chain PO/invoice checks.
- **Government licensing:** Platform-as-a-service for state trade facilitation bodies.

Market Opportunity

India's merchandise exports totalled $437B in FY2024, with MSMEs contributing approximately 45% ($197B). The cross-border trade documentation market in India is estimated at $2–3B annually. MerkNet's addressable segment is precision manufacturing exports — the corridor where Merkantis already operates. At an average transaction fee of ₹5,000 per order across tens of thousands of annual cross-border manufacturing orders, the near-term revenue opportunity is $5–15M, scaling with corridor expansion and financing channel adoption.

Key Partnerships

- ICEGATE / NIC — customs electronic filing
- Indian banks (SBI, ICICI) — financing channel, orderer nodes
- CODISSIA — supplier onboarding in Coimbatore
- Swiss-Indian Chamber of Commerce — corridor validation
- CII / FICCI — pan-India rollout

Risks and Mitigations

Risk	Severity	Mitigation
Government node commitment	High	Pilot designed to run without government nodes (Merkantis simulates roles). Value proposition demonstrated first; government engagement follows.
Supplier adoption	Medium	Zero-infrastructure managed identity removes technology barrier. QR-code interface for factory-floor users. Onboarding via industry associations (CODISSIA, CII).
Fabric 3.0 SmartBFT maturity	Medium	SmartBFT is new in production. Fallback: deploy with Raft (CFT) for Phase 1 where Merkantis controls all orderers; upgrade to SmartBFT when government orderers join.
Bank partnership for financing channel	Medium	Financing channel is Phase 2. Trade channel delivers standalone value. Bank engagement begins in parallel with MVP.
Regulatory change	Low	Architecture is modular; compliance rules stored on-chain and updatable without chaincode changes. DSC compatibility ensures alignment with any future Indian PKI evolution.
GDPR key destruction acceptance	Low	EDPB Guidelines 02/2025 acknowledge this approach. Architecture also supports full off-chain data deletion. Legal counsel engaged for controller/processor agreements.
Competitive entry by well-funded player	Low	Merkantis's 200+ operational engagements and supplier relationships are a defensible moat that cannot be replicated without years of on-the-ground work in the same trade corridors.

Future Roadmap

- **IoT:** Factory sensors with own Fabric CA identities; architecture ready, depends on factory digitization.
- **Portable credentials:** Supplier certifications as Verifiable Credentials via Indy/Aries for cross-network reputation.
- **Advanced finance:** LCs, bank guarantees, multi-bank syndication as on-chain workflows.
- **ML compliance:** Anomaly detection for improbable quality claims as transaction history accumulates.

Signature and Team Information

[To be completed]

Appendix A: Technical Architecture

Network Topology

Ordering Service (SmartBFT)

Shared consensus layer. Orderer nodes sequence transactions into blocks but do not store ledger data or execute chaincode — they are blind relays. Phase 1: Merkantis (2) + DGFT (1) + Customs (1) = 4, F=1. Phase 2: 7 orderers, F=2. Phase 3: further distribution.

Peer Nodes

- **Merkantis:** 2 peers (HA) + FireFly + IPFS.
- **Government:** 1 peer + FireFly. Officers authenticate with DSCs via external CA support.
- **Large suppliers/buyers (Phase 2+):** 1 peer + optional FireFly.
- **Small suppliers/buyers:** Managed identity via Merkantis.

Channel Architecture

Network Channel

All participants. Self-contained in Fabric ledger (no IPFS). Stores: identity registry (including DSC-linked public keys), reputation, compliance rules, milestone history (hashes only). Read-heavy.

Trade Channel

Active traders. All deal lifecycle transactions. Ledger stores only hashes and encrypted blobs. Privacy via PDCs + envelope encryption.

Financing Channel

Banks + Merkantis. Loan lifecycle. Banks verify by hashing against network channel. Separated from trade data.

Privacy Layer

Single PDC “tradedata” on trade channel. Envelope encryption: deal creator generates AES-256 key, encrypts data in PDC and documents on IPFS. Key distributed per-member via ECIES. Public ledger: deal ID, hash, CIDs, status, members only.

UCANs: capability tokens issued to deal parties with delegation chains (e.g., supplier → inspector). No central auth server. Pinned content = hot; unpinned = cold but accessible.

Identity and Access Control

Fabric CA issues X.509 certificates with role attributes (ABAC). Government participants with existing DSCs — issued by CCA-accredited authorities (eMudhra, Sify, NCode) under the IT Act 2000 — use them directly via Fabric's external CA support in MSP configuration. This means a DGFT officer's existing DSC becomes their transaction-signing identity on MerkNet without issuing new credentials.

Certificate attributes: role (buyer, supplier, inspector, customs_officer, etc.), org (MSP ID), certifications (ISO9001-auditor, DGFT-authorized, etc.).

Access control: SubmitRFQ = role=buyer; RecordInspection = inspector + certifications; FileCustomsDeclaration = customs_officer; ConfirmDelivery = buyer + deal PDC membership.

Endorsement: PO (buyer + supplier + Merkantis), inspection (inspector + buyer), customs (Merkantis + customs), reputation (Merkantis only).

Data Flow: Pre-Order

1. Buyer onboards → managed identity on trade channel.
2. RFQ submitted → encrypted in PDC (Buyer + Merkantis), hash on public ledger.
3. Merkantis contacts suppliers → each gets RFQ in own PDC entry.
4. Suppliers respond → encrypted in PDC.
5. Merkantis aggregates, sends options to buyer.
6. Buyer selects → deal PDC created, deal key generated and distributed via envelope encryption.

Data Flow: Post-Order

1. Negotiation → each revision = Fabric transaction; specs encrypted on IPFS.
2. PO recorded → payment terms activated; PO hash to network channel.
3. Manufacturing checkpoints → inspector scans QR, enters data, recorded on Fabric with inspector's signature.
4. Pre-shipment inspection → report + photos encrypted on IPFS, hash on Fabric.
5. Shipping → e-BL minted as token; customs added to deal PDC; CHA pre-populates forms via QR; DGFT officer signs CoO with DSC on-chain; compliance checks run.
6. Delivery → buyer confirms; hash to network channel.
7. Payment tracking → chaincode events; FireFly notifies buyer; late payments on public ledger.
8. Completion → reputation update aggregating payment, quality, delivery metrics.

Deployment

All components containerized (Docker/Kubernetes). Standardized templates. Self-sovereign participants deploy on own infra or Merkantis-provisioned cloud instance.

Appendix B: Competitive Landscape

TradeLens (Maersk/IBM)

Shut down November 2022. Shipping/logistics only. Required participants to run own infrastructure; competitors wouldn't join a Maersk-controlled network. No integration with national PKI or government signature infrastructure.

Marco Polo (R3 Corda)

Shut down late 2023. Bank-to-bank trade finance only. No document lifecycle or manufacturing integration. Never bridged trade execution and finance.

CargoX

Active. Egyptian customs (NAFEZA) for e-BL. Covers e-BL only. Built on Ethereum — crypto-adjacency concerns with Indian regulators.

Indian Government Pilots

NIC/NITI Aayog small-scale PoCs. Single-department, narrow scope, not multi-stakeholder. None integrated DSC-based signing with full trade document lifecycle.

Raymond x Spydra

Product identity on Fabric, 15M assets tokenized. QR-based ownership tracking. Product authentication, not cross-border trade documentation.

Appendix C: Technology Selection Rationale

Why Fabric 3.0+ (Not 2.x)

Fabric 2.x uses Raft — crash fault tolerant only. MerkNet's orderers span Merkantis, DGFT, Customs — independent entities. SmartBFT (3.0+) provides the Byzantine fault tolerance required. Fallback: deploy with Raft for Phase 1 (Merkantis-only orderers), upgrade when government orderers join.

Why Fabric (Not Alternatives)

- **Hyperledger Indy:** Considered for VCs/ZKPs. Fabric CA with ABAC covers all identity needs — parties in a deal know each other, ZKP selective disclosure is unnecessary. DSC support via external CA is more practically valuable than Indy's DID model for Indian government use. Running a separate Indy ledger adds complexity without proportional benefit. Future roadmap item for portable credentials.
- **R3 Corda:** Financial services focus. No multi-channel architecture, no native middleware, no ABAC.
- **Besu / Polygon / Ethereum L2s:** Crypto-adjacency risk with Indian regulators where non-crypto is explicitly preferred.
- **Sawtooth:** Deprecated. Inactive community.
- **Cosmos SDK:** Crypto-adjacent ecosystem. Overkill for permissioned deployment.

Appendix D: Smart Contract Suite

Seven chaincode packages across three channels. Multi-party signature thresholds enforced via ThresholdPolicy chaincode on each org's peer, with configs in implicit private collections.

1. DealLifecycle (Trade Channel)

States: RFQ_SUBMITTED → QUOTES_RECEIVED → SUPPLIER_SELECTED → NEGOTIATION → PO_CONFIRMED → IN_PRODUCTION → SHIPPED → DELIVERED → COMPLETED | DISPUTED. PO confirmation: buyer + supplier + Merkantis endorsement; ThresholdPolicy if value exceeds org limit.

2. QualityVerification (Trade Channel)

SCHEDULED → IN_PROGRESS → REPORT_SUBMITTED → APPROVED | REJECTED | REWORK. Inspector signs report with their credential; signature verifiable by buyer's peer. Approval emits event to PaymentTracking.

3. DocumentManagement (Trade Channel)

Documents as tokens via FireFly: DRAFTED → ISSUED → TRANSFERRED → RETIRED. e-BLs as NFTs with endorsement chain in metadata. CoO issued after compliance validation and signed by DGFT officer's DSC — signature cryptographically bound to document hash. ABAC-enforced.

4. PaymentTracking (Trade Channel)

TERMS_DEFINED → MILESTONE_DUE → PAYMENT_RECORDED → LATE_PAYMENT_FLAGGED → SETTLED. No funds on-chain. Late payments on public ledger, feeding reputation.

5. Compliance (Trade Channel)

Validates: DGFT export policy (IEC, HS codes), destination regulations (CE, REACH, RoHS), bilateral agreements, sanctions. Invoked before CoO/customs issuance. Failures block with documented reason. Rules on network channel, updatable without chaincode upgrade.

6. Reputation (Network Channel)

Milestone hashes: PO_CONFIRMED, DELIVERY_COMPLETE, INVOICE_PAID (no commercial details). Aggregates: payment timeliness, quality acceptance, delivery reliability, communication responsiveness. Scores visible to all members, determine access tier.

7. SupplyChainFinance (Financing Channel)

OCEN-aligned. FINANCING_REQUESTED → VERIFIED (bank hashes doc, checks network channel) → TERMS_OFFERED → TERMS_ACCEPTED → DISBURSED → REPAID | DEFAULTED. Loan lifecycle independent of trade channel. Merkantis's LSP positioning requires bank partnerships and regulatory alignment that are part of Phase 2 roadmap.

Appendix E: Full Regulatory Framework

Indian Regulations

IT Act, 2000 (amended 2008)

Electronic records (S.4), digital signatures (S.5), evidence admissibility (S.65B Indian Evidence Act). DSCs issued under this Act by CCA-accredited authorities are the identity and signing foundation for MerkNet. Fabric transactions signed with DSCs carry the same legal weight as DSC-signed documents under the Act.

Bills of Lading Act, 2025

Enacted August 2025. Full legal recognition of e-BLs. Central Government to notify technical rules; MerkNet architecturally ready.

Carriage of Goods by Sea Act, 2025

Enacted August 2025. Aligns India with Hague-Visby Rules. Modernizes carrier liability and multimodal transport.

Foreign Trade Policy 2025

Explicitly includes blockchain security for trade records.

Customs Act / ICEGATE

Integration via FireFly REST API. Compliance chaincode validates before submission. QR pre-population for CHAs.

FEMA

Financing channel within RBI guidelines for short-term trade credit.

RBI Digital Lending Directions, 2025

Mandatory Key Fact Statements, direct disbursement, LSP fee transparency. MerkNet's financing channel is designed to comply; actual LSP status requires regulatory engagement in Phase 2.

TReDS

Complementary: TReDS for receivables, MerkNet for PO-based pre-shipment finance.

OCEN 4.0

Merkantis intends to position as LSP. APIs designed for OCEN consent management, loan origination, disbursement flows. Bank partnerships required for activation.

DPDP Act, 2023

Data minimization (hashes only on ledger), consent at application layer, key destruction as erasure, cross-border via blacklist approach.

International Regulations

GDPR

Extraterritorial. On-chain: salted hashes (EDPB Guidelines 02/2025 acknowledge these as personal data under GDPR, but note that if off-chain records are deleted and the hash becomes "disconnected," this may satisfy right to erasure — an evolving regulatory position). Off-chain: encrypted PII. Key destruction renders ciphertext permanently unreadable. Clear controller/processor agreements.

UNCITRAL MLETR

India not ratified. Bills of Lading Act 2025 achieves similar outcomes. e-BL follows MLETR principles.

ICC / WCO / WTO TFA

Document schemas aligned. Digital single window per Article 10.

Appendix F: Security, Scalability & Performance

Security Architecture

- **SmartBFT:** Tolerates F Byzantine nodes (3F+1 orderers). Phase 1: F=1, Phase 2: F=2.
- **DSC identity:** Government officers sign with existing DSCs. Any peer can verify signatures cryptographically. No manual verification.
- **mTLS:** All communication mutually authenticated.
- **Execute-order-validate:** Orderers never see plaintext.
- **Key management:** Orderer/CA root keys HSM-backed (CloudHSM or YubiHSM). Peer keys encrypted-at-rest, upgradeable.
- **Chaincode isolation:** Isolated Docker containers.
- **On-prem option:** Locked-down Linux appliance with integrated HSM for participants requiring on-premises.

Scalability

Metric	Estimate
Transactions per order	~100–300
Concurrent orders (Phase 1)	~50–100
Concurrent orders (Phase 2)	~500–1,000
Peak TPS required	~5–10
Fabric SmartBFT capacity	~3,000 TPS
Headroom at Phase 2	~300–600x
Ledger growth / 1,000 orders	~1–5 GB
IPFS storage / 1,000 orders	~50–100 GB

Resource Requirements

Participant	Compute	RAM	Storage	Cost/Mo
Merkantis (MVP)	4 cores	16 GB	200 GB SSD	$80–150
Merkantis (Growth)	12 cores	32 GB	500 GB SSD	$250–400

Government node	4 cores	16 GB	200 GB SSD	$80–150
Bank node	4 cores	16 GB	200 GB SSD	$80–150
Self-sovereign	2 cores	8 GB	100 GB SSD	$40–80
Managed identity	—	—	—	$0

Performance

Fabric 3.0 SmartBFT: ~3,000 TPS. A cross-border order: ~100–300 transactions over weeks. At 1,000 concurrent orders, peak demand ~5–10 TPS. Document/QR verification: sub-second. IPFS pinned retrieval: < 5 seconds.